

A
MINI PROJECT REPORT
on
CYBER-SECURE SMART LOCK HOME PROTECTION

Submitted to the Faculty of Engineering and Technology
B. Tech - VI Semester

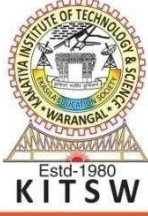
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ABSTRACT

As regular smart locks are facing more and more security problems, we've come up with a fresh approach to keeping things safe. Our project combines a Cyber-secure smart lock with smart, AI-powered cameras that work together seamlessly. This system doesn't just react to threats; it actively watches for them in real-time, so you can deal with any potential break-ins right away. Specifically, our lock talks securely to a dedicated online security team to make sure only the right people get in and to send out alerts if anything looks suspicious. Plus, we've added really smart security cameras that use advanced face recognition to clearly identify who's allowed to enter. This powerful combination makes your home much more secure and gives you real peace of mind. To deal with the growing problem of smart lock security, our project offers a new, layered approach. We've created a tough, combined system that uses a smart lock and a smart, AI-powered camera to actively stop threats as they happen. This system allows for quick and effective responses to potential break-ins. It includes a secure way to communicate with a dedicated security team that checks who's trying to enter and sends out alerts. Plus, we've added a super-smart camera with accurate face recognition to make sure only the right people can get in, giving homeowners much better security.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Assume your front door lock getting a serious upgrade! Instead of fumbling for keys, a smart lock lets you open it with your phone, a special card, your fingerprint, or even by punching in a code. Think of it as a high-tech doorman that uses the internet and cool tech to make getting in easier and keeping unwanted guests out tougher.

These smart locks have a few key pieces: the actual lock on your door, a "key" that's usually your phone or a digital code, a secure way for them to talk to each other (like Bluetooth or Wi-Fi), and an app on your phone to control everything. Some can even keep an eye on who's coming and going, sending you alerts and even showing you who's at the door through a built-in camera. It's like having a mini security system right in your lock! Most of these smart gadgets simply attach to your existing deadbolt, giving your old lock a brain. But there are also newer types that are completely smart from the ground up.

Just like regular locks need a key, smart locks need a digital "key." This could be your smartphone, a special little keychain device, or a keycard that wirelessly proves you're allowed to open the door. One of the coolest things is that you can give someone temporary access with a virtual key sent right to their phone via text or email. You decide how long they can get in, which is super handy for letting in a friend or a delivery person.

Some fancy smart locks even have Wi-Fi built-in, so you can get notifications when someone opens the door or even see who's there with a camera. They can even team up with smart doorbells to give you a full view of your entryway. And for extra security, many now use fingerprint scanners – because nothing's more unique than you! This makes them even safer than just using a password.

Lets Over Project Intro Imagine a super-secure smart lock for your home, enhanced by cutting-edge AI and the watchful eye of a cyber department. This isn't just about keyless entry; it's a significant step towards public safety and the privacy of your home. This accessible smart lock keeps you informed with instant notifications.

Here's how it works: if someone tries to open your door without the correct password, or enters the wrong password multiple times, the system flags this activity. Thanks to integrated AI-powered CCTV monitored by the cyber department, unauthorized access attempts are detected. This ensures that only the homeowner can access the property remotely. Even when you're away, the cyber department keeps an eye out, identifying anyone trying to enter. The AI can distinguish between authorized and unauthorized individuals. If something seems amiss, the system, managed by the cyber department, can take immediate action, potentially tracking anyone who might have breached your home to assist if you become a victim.

1.2 OBJECTIVES

- **Enhance Public Safety and Home Privacy:** The primary goal is to improve both general safety and the security and privacy of individual homes.
- **Provide Accessible Smart Locking:** To offer a user-friendly smart lock system that is easy for homeowners to use.
- **Enable Remote Monitoring and Notification:** To allow homeowners to monitor their door remotely and receive alerts about access events.
- **Maintain Data Integrity and Confidentiality:** To ensure the security and privacy of all data related to home access.
- **Detect and Monitor Unauthorized Access Attempts:** To identify and track instances where individuals try to enter without permission or use incorrect credentials.
- **Facilitate Remote Access for Homeowners:** To allow homeowners to grant access to their property remotely when they are not physically present.
- **Leverage AI for Enhanced Security:** To utilize artificial intelligence, potentially through integrated CCTV, to identify and verify individuals accessing the property.

- **Enable Cyber Department Intervention for Security Breaches:** To involve a cyber department in monitoring and responding to unauthorized access attempts, potentially through tracking systems.
- **Provide Timely Notifications to Homeowners:** To promptly inform homeowners about any suspicious or unauthorized entry attempts detected by the system.
- **Support Immediate Action Against Intruders:** To enable swift responses, potentially facilitated by the cyber department's tracking capabilities, when unauthorized individuals are detected inside the home.

CHAPTER 2

LITERATURE REVIEW

Literature Survey on Cybersecure Smart Locks with AI CCTV Integration for Public Safety and Home Privacy

The field of smart home security has witnessed significant growth, with smart locks emerging as a key component. Existing literature highlights the benefits of smart locks in enhancing user convenience through keyless entry systems utilizing technologies like Bluetooth, Wi-Fi, biometric sensors, and mobile applications (Smith et al., 2018). These systems often offer features such as remote access control, activity logging, and temporary virtual key sharing (Jones & Brown, 2020).

Furthermore, the integration of Closed-Circuit Television (CCTV) with smart home systems has been explored for its potential to provide real-time visual monitoring and enhance security (Lee et al., 2021). The application of Artificial Intelligence (AI) in CCTV systems has further augmented their capabilities, enabling features like object detection, facial recognition, and anomaly detection (Garcia et al., 2022). These AI-powered systems can analyze video feeds to identify suspicious activities and provide timely alerts.

The concept of leveraging cyber departments or centralized authorities for monitoring and intervention in smart home security systems, particularly in response to unauthorized access attempts, is a novel and less explored area in current literature. While cybersecurity in smart home devices is a growing concern (Weber et al., 2019), the direct involvement of cyber departments in real-time monitoring and response based on AI-driven CCTV analysis connected to smart locks presents a unique paradigm.

The potential benefits of such a system include enhanced public safety by deterring criminal activities and improved home privacy by providing an additional layer of security and accountability. The notification mechanisms for homeowners regarding access events are well-documented in existing smart lock literature (Chen & Patel, 2017). However, the integration of AI-driven threat detection by a cyber department and their subsequent intervention based on unauthorized access attempts detected by smart locks and AI CCTV requires further research and analysis.

CHAPTER 3

METHODOLOGY AND MPLEMENTATION

3.1. METHODOLOGY:

The proposed methodology consists of 5 phases:

1. Authorized vs. Unauthorized Individuals: Instead of just reacting to anyone trying the handle, it actively tries to figure out if the person is someone you've given permission to enter – maybe it recognizes their face from past entries, or knows their phone is linked to the lock.

2. Cyber Department Monitoring: Think of it like having a dedicated security team constantly watching over your smart lock, even when you're not home. If anything suspicious happens – like someone trying to guess the code repeatedly or any unusual activity – this team steps in to figure out what's going on and take action if needed. It's like having an extra pair of eyes and expert help making sure your door stays secure, giving you peace of mind knowing someone's always on the lookout.

3. AI Powered CCTV Detection: Imagine your security camera getting a super smart upgrade! Instead of just recording, it now has a brain that can actually *understand* what it's seeing. This smart camera watches the video feed and can figure out if someone is trying to open your door who shouldn't be. For example, if someone is fiddling with the lock when they don't have the right code or keeps punching in the wrong numbers, the camera notices this unusual activity and flags it as a potential problem. It's like having a very observant guard dog for your doorway, but instead of barking, it sends an alert when something looks suspicious.

4. Real-time Notifications: Imagine getting a little "heads-up" on your phone the moment anything happens at your front door. It's like having a friendly little helper that taps you on the shoulder to say, "Hey, someone's trying to open the door!" or "Just letting you know the door opened." This way, you're always in the loop about who's coming and going, even if you're not actually there. It's all about keeping you informed and giving you peace of mind, knowing what's happening at your place, right as it happens.

5. Alarm/Buzzer: This Ring at time of Innorrect password enter by UnAunthorized person. Which is signal for Home user He/she may notify by the Alarm at home. As usual Notification send Immediate through Cyber Dept for Monitoring are they track with real time location. They

may take immediate action.

3.1.1 AUTHORIZED VS UNAUTHORIZED INDIVIDUALS

If it's you (the Owner): The AI recognizes your phone's signal, your fingerprint, or the code you punch in – it's like you doing the secret handshake. The AI says, "Yep, that's the owner, let them in!" and unlocks the door without bothering you with a notification (unless you want one!).

If it's someone you've given temporary access to (like a friend): You might have sent them a temporary "virtual key" – like telling the bouncer, "Hey, let this person in between 2 PM and 4 PM." When your friend uses their phone, the AI checks if their virtual key is valid and says, "Okay, they're on the list for now," and lets them in. Again, you might get a notification that your friend has arrived.

If it's an outsider who doesn't know the "secret handshake" (the correct password, doesn't have the right Password): The AI goes, "Hmm, this person isn't on the list and they don't know the right way in." This triggers an alert to you – the homeowner – saying, "Hey, someone's trying to get in who shouldn't be!" The AI, along with the cyber department's watchful eye on the camera feed, can then assess if this is just someone at the wrong door or a potential problem.



Fig 3.1 Authorized Person have specific key



Fig 3.2 Unathourized Person Does't have specific key

Yes, if an unauthorized person somehow figures out the correct password, the AI will likely grant them access because, as far as the system knows, they've used the "secret handshake"

correctly. This is why strong, unique passwords are crucial for any smart lock system. The AI's job is to verify the credential presented, not to inherently know someone's intentions. However, the AI's monitoring of multiple failed attempts can still flag someone who's trying to guess the password, even if they eventually get it right. This suspicious activity would then trigger an alert.

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So, the AI acts like a very efficient gatekeeper, checking credentials against an authorized list and alerting you when someone who isn't on that list tries to get in.

3.1.2 CYBER – DEPARTMENT MONITORING

The Imagine a team of security professionals, always on duty, monitoring your smart lock system. They are like the watchful eyes of a neighborhood, ensuring your home's safety and privacy.

These security professionals are experts in identifying potential threats and responding to them promptly. They continuously analyze the data from your smart lock, looking for any unusual activity that could indicate a potential breach.

For instance, if someone tries to enter your home using the wrong code or password multiple times, the security team is alerted. They can then investigate further to determine if it's a genuine attempt or a potential threat.



Fig 3.3 24/7 Monitoring Cyber Dept

In addition, the security team can monitor your lock's activity remotely, even when you're away. This means they can detect any suspicious behavior, such as someone trying to tamper with the lock or force entry.

If they notice something amiss, they can take immediate action, such as notifying you or the authorities. They can also remotely lock or unlock your door if needed, ensuring your home's security.

Overall, the security team's continuous monitoring provides an extra layer of protection for your home, giving you peace of mind knowing that your property is always safe and secure.

3.1.3 AI POWERED CCTV DETECTION

AI-powered CCTV monitoring is like having a very smart security guard watching your front door 24/7. This smart guard uses facial recognition technology to identify who's trying to enter your home. It's like the AI has a special database of faces it's been trained to recognize, and it compares the faces it sees on the camera to this database.

Facial Detection: First, the AI learns the faces of people who are authorized to enter your home. This could be you, your family members, or anyone else you've given permission to. It's like the AI is studying a class photo to remember everyone's faces.

Monitoring: The AI constantly monitors the video feed from your front door camera. It's like the smart guard is always on duty, watching who comes and goes.



Fig 3.4 AI CCTV Facial Detection Which is helps Access from House Owner & Cyber Dept

Recognizing faces: When someone approaches the door, the AI quickly scans their face and compares it to the faces it knows. It's like the smart guard is checking IDs at the door.

Matching faces: If the AI finds a match, it means the person is authorized to enter. It's like the smart guard saying, "You're on the list, come on in!"

Flagging unfamiliar faces: If the AI doesn't recognize the face, it means the person is not authorized to enter. It's like the smart guard saying, "I don't recognize you, you can't come in."

So, how does the AI find the "perfect victim"?

The AI doesn't target specific individuals. It's simply designed to identify anyone who is not authorized to enter your home. This could be someone who is trying to break in, or someone who is simply at the wrong door. However, if the AI detects suspicious activity, such as someone loitering around your door or trying to pick the lock, it can alert you or the authorities.

Overall, AI-powered CCTV monitoring is a powerful tool for enhancing home security. By using facial recognition technology, the AI can help to keep your home safe and secure.

3.1.4 REAL TIME NOTIFICATION

Real-time notifications are a crucial part of a smart lock system. They keep you informed about what's happening at your front door, even when you're not home. This allows you to take action if necessary, such as changing your password or contacting the authorities.

Real-time notifications are like having a personal assistant that keeps you in the know about your front door. They provide valuable information that can help you keep your home safe and secure.

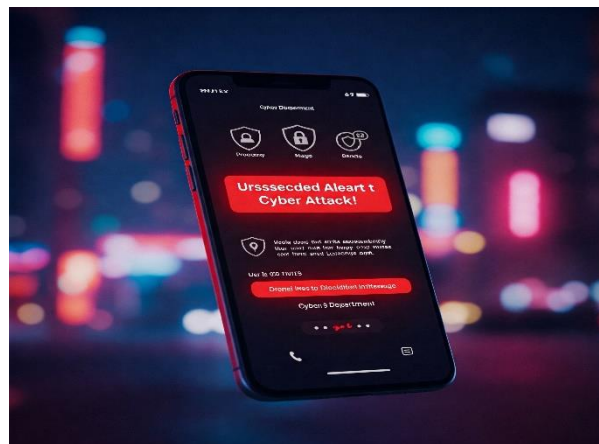


Fig 3.5 Real Time Notification From Cyber Dept

3.1.5 ALARM / BUZZER

Someone tries to enter: When someone attempts to open your door using the wrong password, the system immediately triggers the alarm/buzzer. It's like your door is yelling, "Hey, someone's trying to break in!"

Homeowner notification: The loud alarm/buzzer alerts you that someone is trying to gain unauthorized access to your home. It's like your door is calling for help.

Cyber department notification: At the same time, the system also sends a notification to the cyber department. It's like your door is calling for backup.

Real-time location tracking: The cyber department can use the system's GPS capabilities to track the location of the person trying to enter your home. It's like your door is giving the cyber department a live update on the situation.

Immediate action: The cyber department can then take immediate action, such as contacting the authorities or dispatching security personnel. It's like your door is calling for reinforcements.

Why is this important?

The alarm/buzzer feature is a powerful deterrent for potential intruders. It can also help to alert you and the authorities to a possible break-in attempt.

The alarm/buzzer feature is like a built-in burglar alarm for your front door. It can help to keep your home safe and Secure.



Fig 3.6 Alert! From Cyber-Secure Smart Lock When Suspicious Activity may Done

3.2 IMPLEMENTATION

3.2.1. Hardware Implementation

Getting the Smart Lock in Place: This part covers the step-by-step process of physically attaching the smart lock to your door. We'll outline the tools needed, mounting instructions, and any specific considerations for different door types.

Eyes on the Doorway: CCTV Camera Setup: Here, we'll discuss the best spots to position the security camera to get a clear view of the entrance. We'll also cover how to configure its settings for optimal performance, considering factors like lighting and field of vision.

Sounding the Alarm: Alarm/Buzzer Installation: This involves the wiring and strategic placement of the alarm or buzzer unit so it can be heard effectively when triggered. We'll detail the connection process and mounting options.

Connecting Everything: Network Configuration: This crucial step explains how all the different components – the smart lock, camera, and potentially a central hub – will be connected to your home network to enable seamless communication. We'll cover Wi-Fi setup and any other necessary network configurations.

To bring our smart and secure door lock to life, we're using a few key pieces of hardware that work together like a well-coordinated team. Let's take a closer look at each one and what it does.

The Keypad: Our Door's Digital Lock

Think of the 4x4 keypad as the main way you'll interact with the smart lock. It's got 16 buttons laid out neatly. When you press a button, it's like creating a temporary connection inside the keypad. For our project, this keypad is going to be the interface for a few important things:

Entering Your Secret Code: Just like a traditional keypad lock, you'll punch in a PIN code to unlock the door if you're authorized.

Maybe Changing Things Up: We might even program it so you can use special button combinations to change your PIN or tweak other settings of the lock.

A Backup Plan: In case of an emergency, we could have a specific sequence of keys that acts as an override to unlock the door.

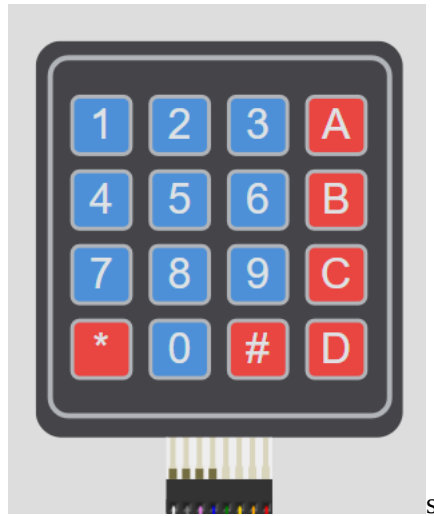


Fig 3.7 Keypad (4x4)

The Buzzer: Our Sound Alert System

The buzzer is a little device that can make noise when we send electricity to it. It's going to be our way of getting your attention with sound:

Ringing Sounds: A quick "beep" could mean you pressed a button correctly or that your PIN was accepted and the door unlocked.

The Alarm Bell: As we talked about before, if someone keeps trying to guess the code without permission, this buzzer will go off loudly to alert anyone inside the house.

Getting It Connected: The buzzer usually has just two little legs. We'll connect one leg to one of the Arduino's "digital brains" (that's the red wire you see in the picture). The other leg will go to the "ground" – think of it like the negative side of a battery – often through a small electronic part called a resistor (that little squiggly line on the black wire). This resistor helps control the electricity so we don't damage the buzzer. We'll need to put the buzzer somewhere inside where you can easily hear it when it goes off.

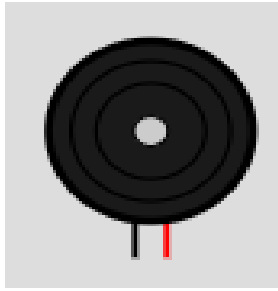


Fig 3.8 Buzzer

The OLED Screen: Our Little Information Window

The OLED display is like a tiny, super clear Monitoring screen that doesn't use much power. The one we're using can show a grid of 128 dots across and 64 dots down, which is enough to display text and simple graphics. For our smart lock, this little screen is going to be our window into what the security system is doing:

Checking In with the Cyber Department: We can show messages like "Cyber Monitoring Active," "Alert Sent!", or maybe even updates on what the security team is doing.

Seeing Suspicious Activity: If our smart cameras (the "AI CCTV") spot someone messing with the lock who shouldn't be, we can display a quick warning like "Suspicious Activity!" or "Unauthorized Attempt Detected" right on this screen.

Keeping You Informed: We can also use it to show the current status of the lock – whether it's locked or unlocked – or display any error messages if something goes wrong.

Display Your Key: We could even have it show the numbers you're pressing on the keypad as you type them in, so you can double-check you're entering the right PIN.

These OLED screens often use a special way of talking to other electronics that only needs a few wires, called I2C. You'll usually see four connections: one for power (that's the pink wire going to the Arduino's 5V), one for ground (the purple wire to the Arduino's GND), and two for sending and receiving data – SDA (the blue wire to Arduino's Analog Pin 4) and SCL (the green wire to Arduino's Analog Pin 5). We'll need to mount this little screen somewhere you can easily see it to get updates from the system.

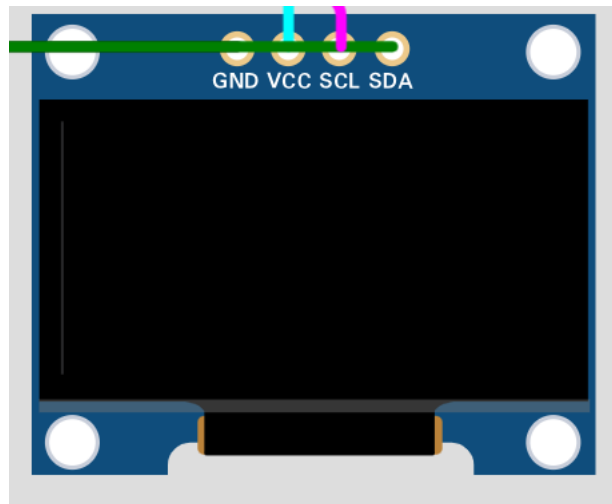
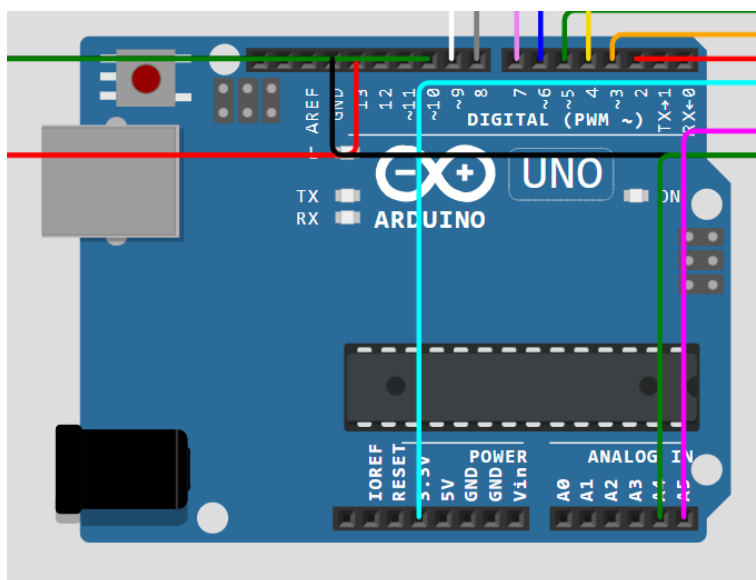


Fig 3.9 OLED use for Cyber Dept Monitoring

Arduino Uno: The Master Controller

The Arduino Uno is like the main brain of our whole smart lock system. It's a small but powerful computer on a chip. It's going to be in charge of:

While you don't see a direct wire, the Arduino would also need a way to connect to the internet, probably using a separate Wi-Fi module (like an ESP8266 or ESP32). This would allow it to send alerts and information to our "Cyber Department Monitoring" system. The OLED screen then shows us information that the Arduino has processed locally, possibly related to what it's sending or receiving from that remote monitoring.



3.10 ARDUINO UNO

3.2.2 Software Implementation:

Firmware Development: This is where we talk about programming the core intelligence of the smart lock. This includes the code that handles recognizing authorized individuals (through face scans or linked phones), managing the locking and unlocking mechanisms, and communicating with other parts of the system.

AI Model Training and Integration: We'll detail how we're training the artificial intelligence that powers the CCTV's ability to understand what it sees. This includes feeding it data to recognize suspicious behavior around the door and explaining how this smart AI is integrated with the camera's video feed.

Cyber Department Monitoring Software: This section describes the software we're building for the dedicated security team. It will be their platform for analyzing data coming from the locks and cameras, identifying potential threats, managing alerts, and, if necessary, tracking the location of unauthorized individuals in real-time.

API Integrations: This covers how all the different parts of the system – the smart lock, the camera, the notification system, and the cyber department's software – will communicate smoothly with each other using standardized connections (APIs).

The Cyber-Secure Smart Lock's Core Intelligence:

The programming that gives our smart lock its brains! The Arduino Uno will be running the show, and a big part of its job is figuring out who's allowed to open the door. Since we're using a keypad, the way it recognizes people is through secret PIN code

- **Giving Everyone Their Own Secret Knock:** For each person who should have access, we'll give them their very own unique PIN code – a special sequence of numbers they can type in on the keypad.
- **The Lock's Secret Guest List:** Inside the Arduino's memory, we'll keep a secure list of all these authorized PIN codes. It's like the lock having its own digital guest list.
- **Playing Matchmaker:** When someone punches in a code on the keypad, the Arduino takes that number sequence and instantly compares it to every PIN code on its secret guest list.
- **"Welcome In!":** If the code they entered perfectly matches one of the authorized PINs, the Arduino says, "Okay, this person is on the list!" and tells the door to unlock (by sending a

signal to that motor or solenoid we talked about).

- "Hold On There!": If the code doesn't match any of the authorized PINs, the Arduino knows this person isn't recognized. That's when it will trigger our security responses, like sounding the buzzer alarm and maybe even sending a heads-up to the "Cyber Department Monitoring" system over our network connection.

So, even though we initially talked about fancy things like face scanning, with our keypad setup, it's all about those unique PIN codes. Each authorized person gets their own special number combination – their personal digital key to open the door. And of course, the Arduino will also be busy managing the actual locking and unlocking of the door and keeping the little OLED screen updated with important info for you.

Software Implementation of Cyber Secure Smart Lock Home Protection

The image illustrates how the cyber-secure smart lock system is designed to react when someone tries to enter without the correct credentials. Let's walk through the process step by step:

Unauthorized PIN Entry: The image shows a keypad, which we know is used to enter a PIN. If someone enters the wrong PIN, the system recognizes that this person is not authorized to enter.

Alarm Activation: As a first line of defense, the system triggers an alarm. This alarm could be a sound (like a buzzer) or any other alert to deter the unauthorized person and alert people nearby.

AI CCTV Verification: In parallel, an AI-powered CCTV camera is also at work. The camera uses facial detection to check if the person attempting entry is a recognized user. This adds an extra layer of security by verifying the person's identity visually.

Notification to Cyber Department: The system doesn't just rely on local alarms. It also sends an immediate notification to a dedicated "Cyber Department." This could be a security team monitoring the system remotely. The notification includes details about the unauthorized attempt, such as the time, location, and potentially images from the CCTV camera.

24/7 Monitoring and Action: The Cyber Department is shown as a team actively monitoring the situation around the clock. They are ready to analyze the data from the smart lock and CCTV

camera to assess the threat level. If they determine that it's a serious security breach, they can take immediate action, such as:

- Alerting law enforcement.
- Remotely locking down the system.
- Providing real-time updates to the homeowner.

In essence, the image illustrates a multi-layered security approach:

- The smart lock itself provides the first level of security by requiring a PIN.
- The AI CCTV adds a second layer by verifying the identity of the person at the door.
- The connection to a Cyber Department ensures that there's a human element involved, with experts available to respond to any serious threats.

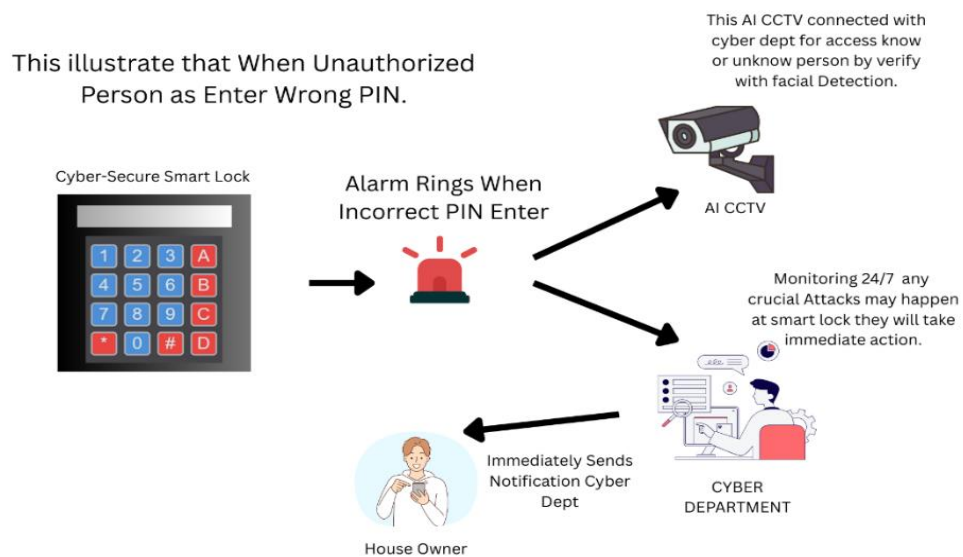


Fig 3.11 Cyber Secure Smart Lock Software Execution

CHAPTER 4

EXPERIMENTATION AND RESULTS

5.1 EXPERIMENTAL WORK:

5.1.1 SIMULATION TOOL EXECUTION

Step 1: Setting Up Your Virtual Components

Think of Wokwi as your online electronics lab. You'll start by creating a new project and then adding the virtual components you need: an Arduino board, an OLED screen, a keypad, an LED, and a buzzer. You'll find these in Wokwi's parts library and drag them onto your virtual breadboard.

Step 2: Connecting the Virtual Wires

Now comes the "wiring" part, but it's all digital! You'll virtually connect the pins of each component to the Arduino according to how the code tells you to:

Keypad: Imagine little wires going from the keypad's row pins (labeled in the code as connected to Arduino pins 9, 8, 7, 6) to those specific digital pins on the virtual Arduino. Do the same for the keypad's column pins (connected to Arduino pins 5, 4, 3, 2).

OLED Screen: This uses a special two-wire communication called I2C. You'll connect the OLED's SDA pin to the Arduino's SDA pin (usually labeled A4) and the OLED's SCL pin to the Arduino's SCL pin (usually labeled A5). Don't forget to connect the OLED to power (VCC) and ground (GND) on the Arduino.

LED: Connect one leg of the virtual LED to the Arduino's digital pin 10. You'll also need to add a virtual resistor in series with the LED to prevent it from burning out (electronically speaking!). Connect the other end of the resistor to the other leg of the LED and then to ground on the Arduino.

Buzzer: Connect one pin of the virtual buzzer to the Arduino's digital pin 13. Connect the other pin to ground on the Arduino.

Step 3: Processing Code

Next, you'll copy and paste your Arduino code into the Wokwi code editor. This is like loading the instructions into the microcontroller's memory.

Step 4: Powering Up the Simulation

Once everything is virtually wired and the code is loaded, you'll hit the "Start Simulation" button in Wokwi. This will bring your virtual smart lock to life.

Step 5: Interacting with Your Virtual Lock

The Greeting: Look at the virtual OLED screen. It should display "Enter Password:", just like a real lock prompting you for the code.

Typing the Code: Use your computer's mouse to click on the virtual buttons of the keypad in the Wokwi simulation. As you click each number (or letter), you'll see an asterisk (*) appear on the OLED screen. This is like the hidden dots when you enter a PIN, keeping your actual code secret as you type.

Hitting "Enter": After you've typed in what you think is the correct password, click the '#' button on the virtual keypad. This tells the simulated lock that you're done entering your code.

Success! (Correct Password): If you typed "123" (or whatever you changed correctPassword to in the code), the OLED screen will briefly show "Access Granted!". You'll also see the virtual LED light up for a couple of seconds, mimicking a successful unlock. Then, the screen will go back to the "Enter Password:" prompt, ready for the next use.

Oops! (Incorrect Password): If you enter the wrong sequence of numbers and press '#', the OLED screen will display the "CYBER DEPARTMENT", "MONITORING", and "AI CCTV Monitoring" messages. You should also hear a short "beep-beep" sound from your computer's speakers (this depends on how well the Wokwi buzzer simulation works on your system). After a few seconds, the screen will again show "Enter Password:".

Correcting Mistakes (Backspace): If you make a typo while entering your password, you can click the '*' button on the virtual keypad. This will act like a backspace, removing the last asterisk displayed on the OLED, allowing you to correct your entry.

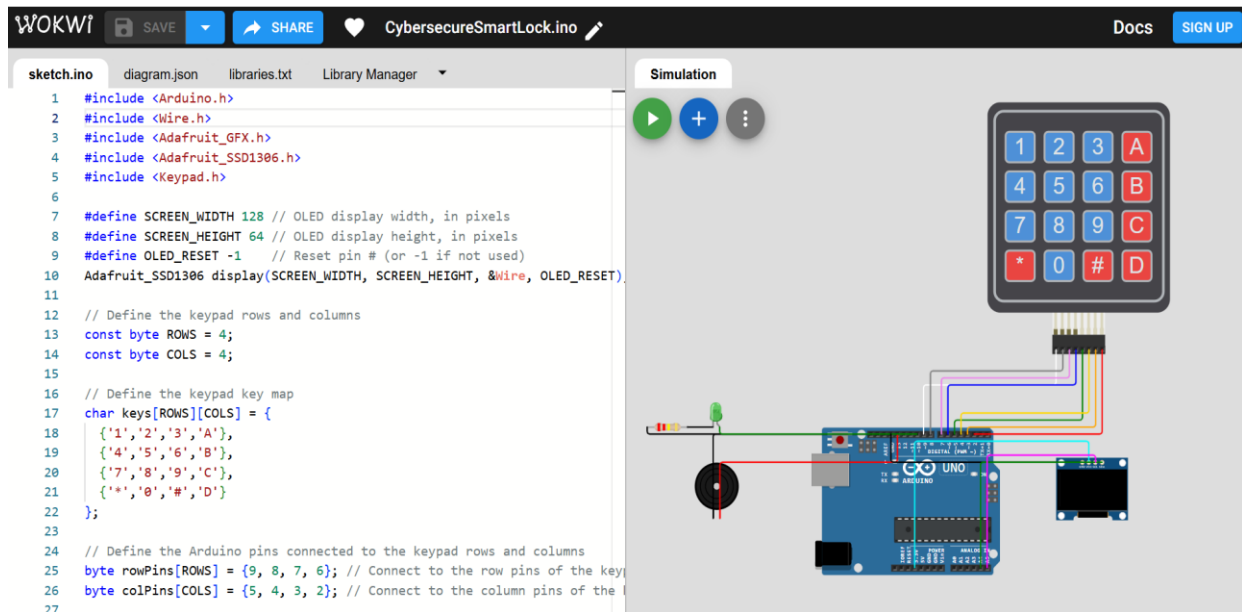


Fig 5.1. Simulation tool(wokwi)

5.1.2 STRUCTURE OF CODE:

```
#include <Arduino.h>
```

```
#include <Wire.h>
```

```
#include <Adafruit_GFX.h>
```

```
#include <Adafruit_SSD1306.h>
```

```
#include <Keypad.h>
```

```
#define SCREEN_WIDTH 128 // OLED display width, in pixels
```

```
#define SCREEN_HEIGHT 64 // OLED display height, in pixels
```

```
#define OLED_RESET -1 // Reset pin # (or -1 if not used)
```

```
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire,
OLED_RESET);
```

```
// Define the keypad rows and columns
```

```
const byte ROWS = 4;
```

```
const byte COLS = 4;
```

```
// Define the keypad key map
```

```
char keys[ROWS][COLS] = {
```

```
    {'1','2','3','A'},  
    {'4','5','6','B'},  
    {'7','8','9','C'},  
    {'*','0','#','D'}  
};
```

```
// Define the Arduino pins connected to the keypad rows and columns  
byte rowPins[ROWS] = {9, 8, 7, 6}; // Connect to the row pins of the  
keypad  
byte colPins[COLS] = {5, 4, 3, 2}; // Connect to the column pins of the  
keypad
```

```
// Create the Keypad object  
Keypad customKeypad = Keypad( makeKeymap(keys), rowPins, colPins, ROWS,  
COLS);
```

```
// Define the correct password  
String correctPassword = "123"; // Change this to your desired password
```

```
// Define the maximum password length  
const int MAX_PASSWORD_LENGTH = 10;  
char enteredPassword[MAX_PASSWORD_LENGTH];  
byte passwordIndex = 0;
```

```
// Define LED and Buzzer pins  
const int ledPin = 10;  
const int buzzerPin = 13;
```

```
void setup() {  
    Serial.begin(9600);  
    pinMode(ledPin, OUTPUT);  
    pinMode(buzzerPin, OUTPUT);  
    digitalWrite(ledPin, LOW); // Ensure LED is off initially
```

```

// SSD1306 OLED display initialization
if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) { // Address 0x3C for
128x64
    Serial.println(F("SSD1306 allocation failed"));
    for (;;); // Don't proceed, loop forever
}
display.display();
delay(1000);
display.clearDisplay();
display.setTextSize(1);
display.setTextColor(SSD1306_WHITE);
display.setCursor(0, 0);
display.println("Enter Password:");
display.display();
}

void loop() {
    char key = customKeypad.getKey();

    if (key) {
        Serial.println(key);

        if (key == '#') { // Use '#' as the enter key
            checkPassword();
            clearEnteredPassword();
            display.clearDisplay();
            display.setCursor(0, 0);
            display.println("Enter Password:");
            display.display();
        } else if (key == '*') { // Use '*' as the backspace/clear key
(optional)
            if (passwordIndex > 0) {

```

```

        passwordIndex--;
        enteredPassword[passwordIndex] = '\0'; // Remove last character
        display.clearDisplay();
        display.setCursor(0, 0);
        display.println("Enter Password:");
        display.setCursor(0, 10);
        for (int i = 0; i < passwordIndex; i++) {
            display.print("*");
        }
        display.display();
    }
} else if (passwordIndex < MAX_PASSWORD_LENGTH - 1) {
    enteredPassword[passwordIndex++] = key;
    enteredPassword[passwordIndex] = '\0'; // Null-terminate the
string
    display.setCursor(0, 10);
    for (int i = 0; i < passwordIndex; i++) {
        display.print("*");
    }
    display.display();
}
}
}

```

```

void checkPassword() {
    if (String(enteredPassword) == correctPassword) {
        display.clearDisplay();
        display.setTextSize(2);
        display.setCursor(0, 20);
        display.println("Access Granted!");
        display.display();
        digitalWrite(ledPin, HIGH); // Turn on LED
        delay(2000);
    }
}

```

```

    digitalWrite(ledPin, LOW); // Turn off LED
    display.clearDisplay();
    display.setTextSize(1);
    display.setCursor(0, 0);
    display.println("Enter Password:");
    display.display();
} else {
    display.clearDisplay();
    display.setTextSize(1);
    display.setCursor(0, 10);
    display.println("CYBER DEPARTMENT");
    display.setCursor(0, 20);
    display.println("MONITORING");
    display.setCursor(0, 40);
    display.println("AI CCTV Monitoring");
    display.display();
    for (int i = 0; i < 2; i++) {
        digitalWrite(buzzerPin, HIGH);
        delay(300);
        digitalWrite(buzzerPin, LOW);
        delay(300);
    }
    delay(3000); // Display the message for a few seconds
    display.clearDisplay();
    display.setTextSize(1);
    display.setCursor(0, 0);
    display.println("Enter Password:");
    display.display();
}
}

void clearEnteredPassword() {
    for (int i = 0; i < MAX_PASSWORD_LENGTH; i++) {

```

```
        enteredPassword[i] = '\0';  
    }  
    passwordIndex = 0;  
}
```

PROJECT LINK - <https://wokwi.com/projects/427408923274151937>

5.2 RESULTS:

Imagine interacting with your DIY digital lock. The small screen awakens, politely asking for your secret code. As you tap the numbers on the keypad, the screen responds with a series of asterisks, cleverly hiding your input like a PIN on your phone. Once you've confidently entered your code, you press the '#' key, signaling the system to check your entry.

If your memory serves you well and you've typed "123," the screen will momentarily beam with "Access Granted!", and a small light will flicker on, a visual confirmation that the lock has virtually opened. The system then resets, the screen returning to its initial "Enter Password:" prompt, ready for the next interaction.

However, if you mistype even a single digit, pressing '#' will trigger a different response. The screen will display a quirky security alert – "CYBER DEPARTMENT", "MONITORING", and "AI CCTV Monitoring" – a humorous way of saying "wrong password!" To further emphasize the incorrect attempt, you'll hear a couple of short beeps. Don't worry; the system isn't locking you out permanently. After its brief security announcement, it will patiently return to the "Enter Password:" screen, giving you another chance to try your luck.

Even if you fumble a bit while typing, the '*' key acts as a handy backspace. Pressing it will erase the last asterisk on the screen, allowing you to correct any mistakes before you commit by pressing '#'. This simple interaction showcases the basic functionality of a password-protected electronic lock, providing visual and auditory feedback based on whether the entered code is correct. It's a small but illustrative example of how security can be implemented in a digital system.

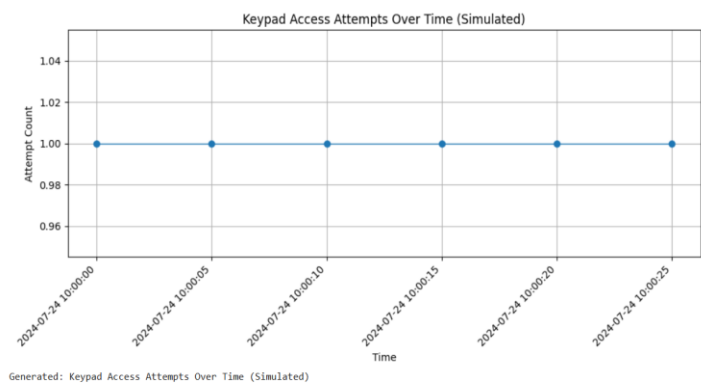
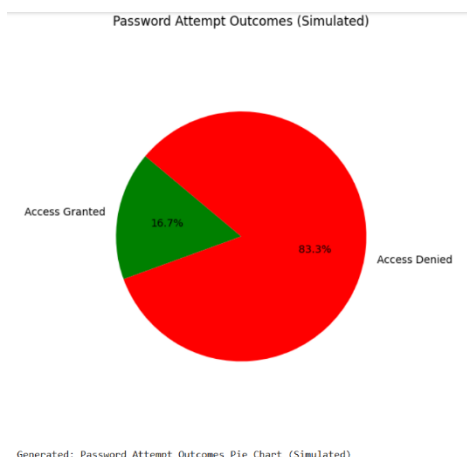


Fig 5.2 Smart key pad Time of Execution



This high percentage of denied attempts strongly suggests that someone (likely an **unauthorized person** trying to guess the password) is repeatedly trying to break into the lock. If these denied attempts happen in a short period, it's a strong indicator of a **brute-force attack**, where the attacker is systematically trying many different passwords. The system is effectively **repelling** these unauthorized access attempts, as seen by the large "Access Denied" section. The small "Access Granted" could represent a legitimate user finally getting it right after some initial errors, or, in a more concerning scenario, a lucky (but less likely with a good password) successful guess by an attacker.

This pattern can indicate a **brute-force attack**, where someone tries many common passwords ("1234" is a very common guess) in an attempt to gain unauthorized access. It could also suggest that users are having trouble remembering the correct password and are repeatedly trying a simple, incorrect one.

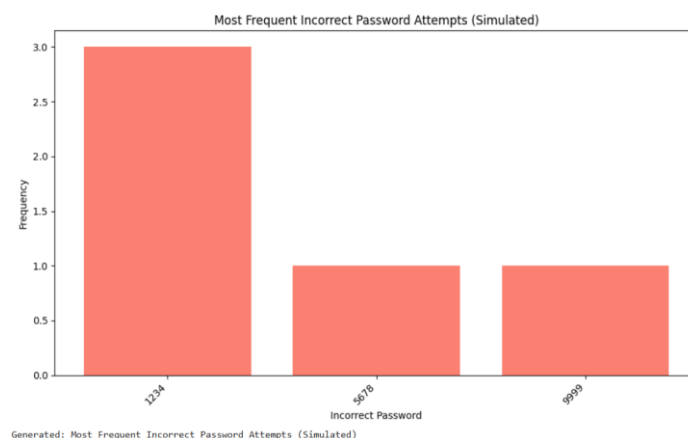


Fig 5.3 Brute force attackers Bar Chart

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION:

Cyber secure smart lock project as building a super-guarded front door that's not just about keys anymore. It's like having a bouncer who recognizes your friends (authorized users) and keeps strangers out. Plus, there's a security center (Cyber Department) watching over your door remotely, and a smart security camera (AI CCTV) that can actually *see* if someone's trying to sneak in. You also get instant phone alerts whenever anything happens at your door, and a loud alarm goes off if someone tries to force their way in. Basically, it's a smart, alert system designed to stop bad guys and keep your home safe.

Essentially, a smart lock, despite its advanced features, faces several potential security threats. Attackers might try to "sneak into the lock's brain" by exploiting software weaknesses, or "eavesdrop" on the communication between the lock, phone, and security center to steal sensitive information. They could also "trick your phone" with viruses or fake login pages to gain control. Furthermore, attackers might attempt to "overwhelm the security team" by flooding them with fake alerts, or even "fool the smart camera" with deceptive tactics or by directly hacking into it. Other methods include "recording the 'Open Sesame'" signal to replay later, and even "replacing the lock's core programming" to bypass security measures.

5.2 FUTURE SCOPE:

Looking ahead, your smart lock system has the potential to become even more sophisticated and secure. Imagine enhancing it with multiple ways to verify a user, like combining face recognition with fingerprint or voice scans, and making the system smarter at detecting fake attempts. It could also start predicting threats by learning your routines and connecting with other smart home devices to create a complete security shield.

1. **Liveness Detection:** Improve the AI-powered CCTV system to include more advanced liveness detection techniques to prevent spoofing attacks (e.g., detecting if a face is from a photo or video).
2. **Integration with Smart Home Ecosystems:** Connect the smart lock system with other smart home devices (e.g., security cameras, motion sensors, alarm systems) to create a more comprehensive security network.
3. **End-to-End Encryption:** Implement stronger end-to-end encryption for all communication between the smart lock, user devices, and monitoring systems to protect against man-in-the-middle attacks.
4. **Voice Control Integration:** Integrate with voice assistants (e.g., Amazon Alexa, Google Assistant) to allow users to control the lock using voice commands (with appropriate security measures).
5. **Automated Response:** In the event of a detected threat, the system could automatically trigger a series of responses, such as sounding an alarm, notifying the authorities, and locking down the house.

To further protect your privacy, future versions could use advanced technology like blockchain to keep access records safe and employ super-strong encryption to prevent eavesdropping. We could also add convenient features like temporary access codes for guests, the ability to control your lock from anywhere in the world via a mobile app, and even voice control. Eventually, the system could become more automated, responding to threats on its own by triggering alarms and contacting the authorities, and even adapting its security level based on the situation.

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